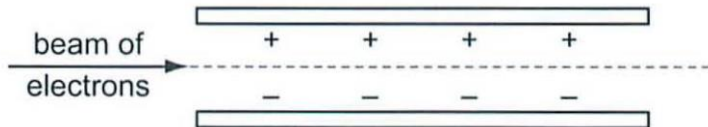


25

A beam of electrons, travelling horizontally towards the right, passes between two horizontal charged parallel metal plates.



There is a vertical electric field between the plates. There is no deflection of the beam because of the presence of a uniform magnetic field in the region between the plates.

In which direction must this magnetic field be?

A

into the paper

B

out of the paper

	C	vertically upwards
	D	vertically downwards
	Answer: A The electric force on the electrons due to the electric field is upwards. Hence the magnetic force must be downwards in order for the beam not to be deflected. Using FLHR, current is towards the left, field must be into the paper.	

26	A magnetic field of uniform flux density 0.045 T is at an angle of 60° below the plane of	
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